conVIZ – Concept Network Visualization

Rahmi Aini

6/7/2023

Background

This package was developed to extend previous methodologies used to analyze co-occurrences of biological concepts (Rusmana et al., 2021; Opfer et al., 2012; Nehm & Ha, 2011) and applies Epistemic Network Analysis (ENA) for educational research. This package is able to identify and automatically visualize the connections among students' conceptual knowledge or any coded data. For example, Fig 1. A) provides a visual representation of the different perspectives on species concepts among high school students. It shows the co-existence of normative ideas (represented by black circles) and naïve ideas (represented by grey circles) of species in high school students' responses.

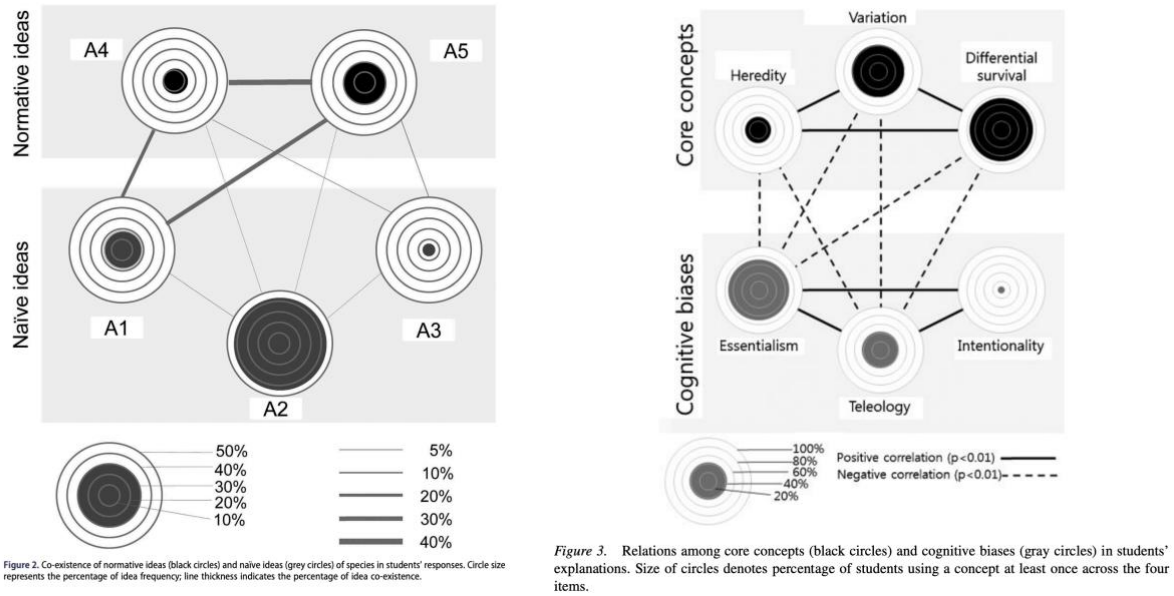


Fig. 1. Visual representation of co-occurrences on A) species concepts among high school students from Rusmana et al. (2020), and B) a correlation structure among students' cognitive biases and core concepts of evolution from Opfer et al. (2012).

ConVIZ, is a R package to calculate descriptive statistics of the coded binary data, generate a document matrix to explore co-occurrence patterns, examine correlations among concepts, and examine the interrelation network between concepts. This package provides valuable insights to educators and researchers, enabling them to understand students' conceptual knowledge and inform instructional strategies.

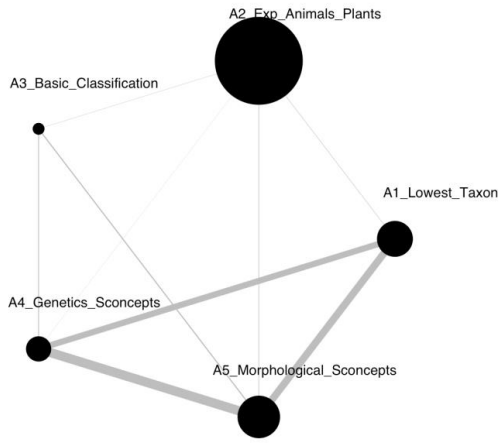


Fig 2. Example of conVIZ output from Rusmana et al. (2020) data

Package Description (Tutorial and Output)

```
library(conVIZ) #Load package
```

#Load example data

Data is a matrix of item responses coded in binary (1 for present; 0 for absent). Missing responses must be declared as "*NA*." Below is the example of dataset from Rusmana et al. (2021).

```
data("data.species") #loads specified datasets
data.species[1:10,] #see example of data
```

Untitled1* | conVIZ Package Information.Rmd | data.species

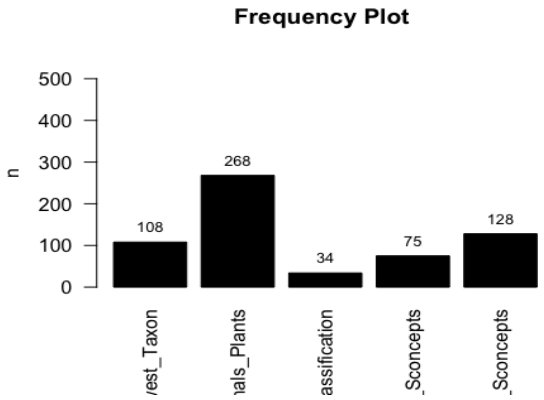
Filter

	A1_Lowest_Taxon	A2_Exp_Animals_Plants	A3_Basic_Classification	A4_Genetics_Sconcepts	A5_Morphological_Sconcepts
1	0	1	0	0	0
2	0	0	0	1	1
3	0	1	0	0	0
4	1	0	0	0	0
5	0	0	0	0	1
6	0	0	0	0	1
7	1	0	0	0	0
8	0	0	0	0	1
9	1	0	0	0	1
10	0	0	0	0	1

The dataset provides examples of five different species concepts: species as the lowest taxon in taxonomy (A1), species as a type of animal or plant (A2), species as the basic classification of living things (A3), genetic species concept (A4), and morphological species concept (A5).

How many of each concept appeared?

```
coocur_freq(data.species)
```



Among the five concepts, concept (A2) was the most frequent among responses.

Do those concepts coexist with other concepts in students' responses?

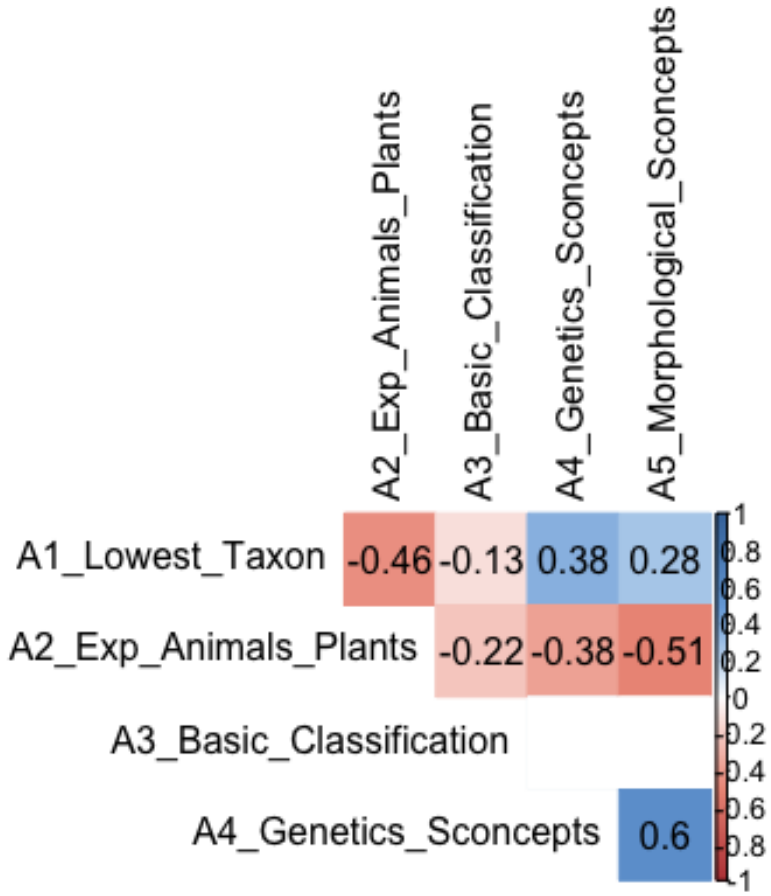
```
coocur_matrix(data.species)
```

	A1_Lowest_Taxon	A2_Exp_Animals_Plants	A3_Basic_Classification	A4_Genetics_Sconcepts	A5_Morphological_Sconcepts
A1_Lowest_Taxon	NA	3	0	43	51
A2_Exp_Animals_Plants	NA	NA	2	1	4
A3_Basic_Classification	NA	NA	NA	6	8
A4_Genetics_Sconcepts	NA	NA	NA	NA	65
A5_Morphological_Sconcepts	NA	NA	NA	NA	NA

The function coocur_matrix creates an adjacency square matrix where the number of rows and columns will be identical. The number of concepts that co-occur with another concept can be seen in the output documents. For instance, a total of 3 students answered concept A2 and concept A1 together in their response, while there are no students that responded to concepts A3 and A1.

Are the concepts showing a statistical correlation to each other?

```
coocur_correl(data.species)
```



The function coocur_correl combines the Pearson correlation test and displays the coefficients in a single visualization. Positive correlations are displayed in blue, and negative correlations are displayed in red. The white box indicates that the correlation is not significant (p >0.005).

How does the co-occurrence between every concept among students look like?

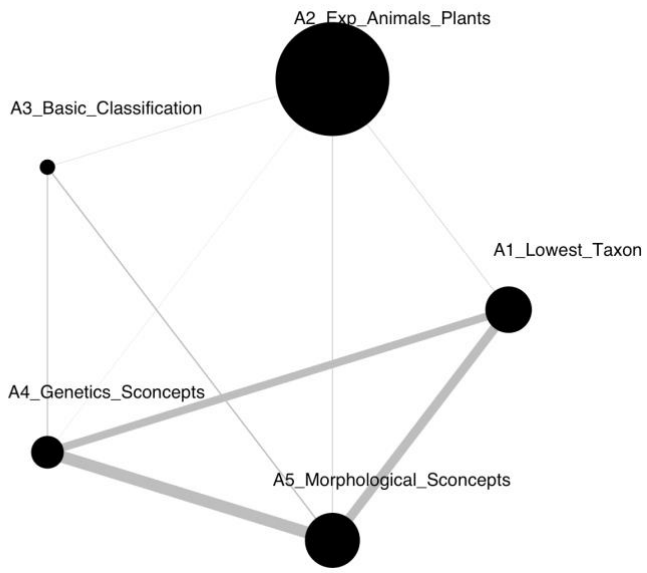
```
coocur_network(data.species)
```

```
##
## Attaching package: 'igraph'
```

```
## The following objects are masked from 'package:stats':
##
##     decompose, spectrum
```

```
## The following object is masked from 'package:base':
##
##     union
```

```
##                             Degree Strength Frequency Percentage
## A1_Lowest_Taxon                  3       97       108       19.7
## A2_Exp_Animals_Plants            4       10       268       48.9
## A3_Basic_Classification          3       16        34        6.2
## A4_Genetics_Sconcepts            4      115        75       13.7
## A5_Morphological_Sconcepts       4      128       128       23.4
```



The function `coocur_network` creates a network plot based on the vertex and edge attributes. These functions are supported by the `igraph` package. The plot is specified in a circle layout (2D) to equalize the distance between vertices.

The vertex (black circle shape) represents one concept**, the edge (gray line connection)** is the co-occurrence between two concepts, and **weight** is the number of co-occurrences between concepts.

Degree indicates the number of connections to a vertex (concept). **Strength** is the sum co-occurrence (edge) connected to a vertex (concept). **Frequency** calculates the number of concepts that occurred in the dataset. See table below for more detail.

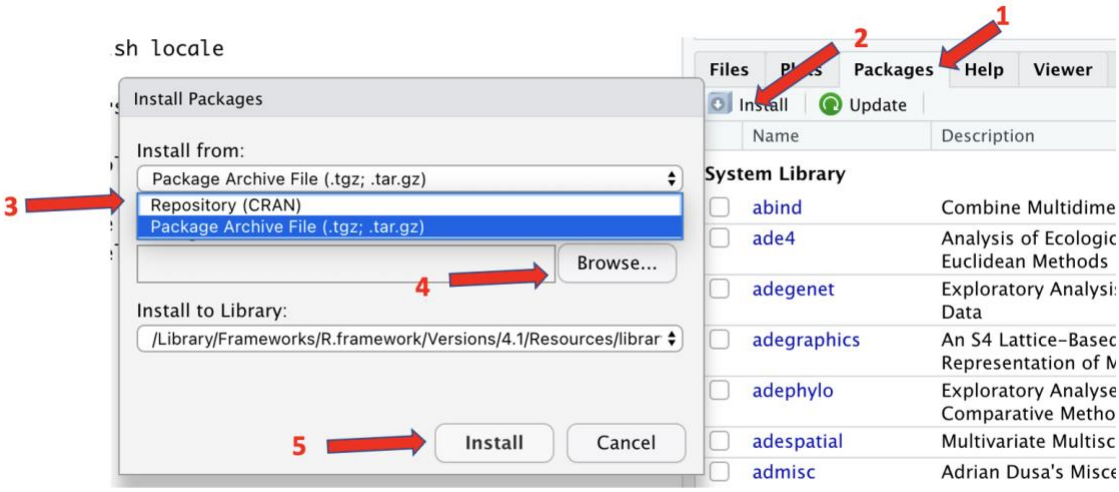
	A1	A2	A3	A4	A5
A1	108	3	0	43	51
A2	3	268	2	1	4
A3	0	2	34	6	8
A4	43	1	6	74	65
A5	51	4	8	65	128
Strength	97	10	16	115	128
Degree	3	4	3	4	4
Frequency	108	268	34	75	128

The size of a vertex (concepts) is standardized based on the sum frequency of the concept divided by the total number of students.

The width of an edge (co-occurrence) is standardized based on the number of co-occurrences between concepts (weight) divided by the sum of all co-occurrences.

The output plot indicates that 268 students responded with the A2 concept, which comprises 49% of the total concepts counted. This concept is also visualized based on the size of the vertex, which is the largest among all concepts. However, this concept has the least co-occurrence with other concepts, indicated by the number of strengths and the width of edges.

Installing conVIZ Package



How to install the package from .tar file in Rstudio:

1. In RStudio: Go to the "Packages" tab in the bottom-right panel.
2. Click on the "Install" button at the top of the "Packages" tab.
3. In the dialog box that appears, click on the "Package Archive File" option.
4. Browse and select the downloaded .tar file of the package.
5. Click on the "Install" button to begin the installation process.

References

Nehm, R. H., & Ha, M. (2011). Item feature effects in evolution assessment. *Journal of Research in Science Teaching*, *48*(3), 237-256.

Opfer, J. E., Nehm, R. H., & Ha, M. (2012). Cognitive foundations for science assessment design: Knowing what students know about evolution. *Journal of Research in Science Teaching*, *49*(6), 744-777.

Rusmana, A. N., Aini, R. Q., Sya'bandari, Y., Ha, M., Shin, S., & Lee, J. K. (2021). Probing high school students' perceptions of the concept of species: a semantic network analysis approach. *Journal of Biological Education*, *55*(5), 472-486.